infrared sensors, and downward-positioning modules to avoid collisions with trees, buildings, and terrain. This allows the drone to intelligently navigate complex environments where manual piloting would require considerable skill.

The impact of ActiveTrack is seen most prominently in film production, action sports recording, tourism, journalism, wildlife documentation, inspection, and personal hobby use. By removing the need for continuous joystick control, ActiveTrack democratizes aerial cinematography, enabling users to capture professional-grade dynamic footage with minimal effort or training. Furthermore, the system has influenced wider drone autonomy trends, serving as a foundational reference for consumer-level object detection, re-identification algorithms, and adaptive flight control behaviors.

While ActiveTrack highlights the benefits of accessible autonomy, it also raises considerations related to privacy, bystander awareness, and responsible use, particularly in public or populated environments. As such, its deployment reinforces the importance of user education, regulatory compliance, and ethical decision-making in everyday drone operations. Overall, DJI ActiveTrack stands as a landmark example of practical, safe, and userfriendly autonomy, bridging the gap between advanced AI technology and everyday civilian drone applications.

Figure 9.3 illustrates the DJI ActiveTrack system, in which a drone autonomously detects, locks onto, and follows a moving subject using computer vision and onboard AI. The drone continuously processes video frames in real time, predicts subject movement, and adjusts flight paths accordingly. Built-in obstacle avoidance sensors ensure safe flight while maintaining camera focus on the target. This system enables smooth